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應用科學學院
Faculdade de Ciências Aplicadas
Faculty of Applied Sciences

Doctor of Philosophy in Computer Applied Technology

Thesis Defence Examination

Thesis Review Comments

Student

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Title of Thesis	Improving the Reliability and Accuracy of Dialogue-Based Natural Language Generation		
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Reviewer

Name	Prof. Catarina Silva	Date	2026-06-08
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Recommendation

☒ Accepted ☐ Suggestion for Revision*

* "Suggestion for Revision" will result in the delay of the oral defence.

Comments

Overview.

The thesis addresses the important and timely problem of improving the reliability and accuracy of dialogue-based natural language generation. It presents a coherent research programme spanning dialogue summarization, source faithfulness, and hallucination mitigation, resulting in several novel methods evaluated across multiple domains and datasets.

Knowledge Base and Literature Review.

The work demonstrates a strong understanding of the state of the art in natural language generation, dialogue summarization, and large language models. The research is well positioned within the existing literature and clearly identifies relevant gaps related to role awareness, discourse structure, and reliability in dialogue-based generation.

Originality and Independence.

The thesis contains several original contributions, including novel frameworks for role-aware dialogue summarization, hierarchical topic modeling, dynamic semantic optimization, and self-supervised faithfulness assessment. The breadth of the contributions and their progressive development suggest a high degree of research independence and scientific maturity.

Arguments and Reasoning.

The research questions are clearly motivated and the methodological choices are well justified. The progression from dialogue summarization to reliability-sensitive generation follows a logical trajectory, and the conclusions are generally supported by the presented empirical evidence.

Evaluation within the Field.

The proposed methods are evaluated on multiple benchmarks spanning different domains and languages, providing convincing evidence of their effectiveness. The inclusion of comparative analyses with contemporary large language models further strengthens the relevance of the work within the field.

Completeness and Coherence.

The thesis is comprehensive and presents a coherent narrative connecting the different research contributions. Individual studies are integrated effectively into a unified research agenda focused on trustworthy dialogue-based generation.

Conclusion.

This thesis represents a substantial contribution to the field of natural language processing, particularly in dialogue-based generation and reliability-aware language modeling. The work advances both methodological development and empirical understanding, while addressing challenges of increasing practical significance.

Additional Comments.